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SY/T 5504.1—2013

中文/English

代替 SY/T 5504.1—2005

油井水泥外加剂评价方法 第 1 部分: 缓凝剂

Evaluation method for well cement additives—
Part 1: Retarder

2013—11—28 发布

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前 言

SY/T 5504《油井水泥外加剂评价方法》分为8个部分：

- 第1部分：缓凝剂；
- 第2部分：降失水剂；
- 第3部分：减阻剂；
- 第4部分：促凝剂；
- 第5部分：防气窜剂；
- 第6部分：减轻剂；
- 第7部分：加重剂；
- 第8部分：膨胀剂。

本部分为SY/T 5504的第1部分。

本部分按照GB/T 1.1—2009《标准化工作导则 第1部分：标准的结构和编写》给出的规则起草。

本部分代替SY/T 5504.1—2005《油井水泥外加剂评价方法 第1部分：缓凝剂》，与SY/T 5504.1—2005相比，主要变化如下：

- 增加了物理性能表，删除了pH值及凝固点的性能要求（见4.1，2005年版的4.1）；
- 增加了应用性能表中稠化线形突变值的要求（见4.2）；
- 增加了稠化时间延长值指标及试验方法（见4.2和5.4.3.3）；
- 删除了稠化线形检查项目（见2005年版的4.2.1）；
- 删除了pH值测定试验方法（见2005年版的5.3.6）；
- 删除了凝固点测定试验方法（见2005年版的5.3.7）；
- 删除了稠化时间可调性指标及试验方法（见2005年版的4.2.1和5.4.3.2）；
- 删除了温度敏感性试验指标及试验方法（见2005年版的4.2.1和5.4.3.6）；
- 删除了加量敏感系数指标及试验方法（见2005年版的4.2.1和5.4.3.7）；
- 增加了加量敏感度指标及试验方法（见4.2和5.4.3.4）；
- 增加了温度敏感度指标及试验方法（见4.2和5.4.3.5）；
- 删除了缓凝剂质量稳定性指标及试验方法（见2005年版的4.2.2和5.5）；
- 修改了代表性温度点的确定方法（见5.4.3.1，2005年版的5.4.3.1）；
- 修改了评价报告（见附录A，2005年版的第6章）。

请注意本文件的某些内容可能涉及专利。本文件的发布机构不承担识别这些专利的责任。

本部分由石油钻井工程专业标准化委员会提出并归口。

本部分起草单位：中海油田服务股份有限公司。

本部分参加起草单位：中国石油化工股份有限公司石油工程技术研究院、中国石油集团海洋工程有限公司天津中油渤星工程科技有限公司。

本部分主要起草人：王永松、宋维凯、王同友、段连超、沈伟、丁士东、沙林浩。

本部分代替了SY/T 5504.1—2005。

SY/T 5504.1—2005的历次版本发布情况为：

- SY/T 5504—1992，SY/T 5504—1996。

本部分以中文和英文两种文字出版。当英文和中文两种版本有歧义时，以中文版本为准。

油井水泥外加剂评价方法

第1部分：缓凝剂

1 范围

SY/T 5504 的本部分规定了油气井注水泥作业用油井水泥缓凝剂的性能要求、评价方法及评价报告。

本部分适用于油气井注水泥作业用油井水泥缓凝剂的评价。

2 规范性引用文件

下列文件对于本文件的应用是必不可少的。凡是注日期的引用文件，仅注日期的版本适用于本文件。凡是不注日期的引用文件，其最新版本（包括所有的修改单）适用于本文件。

GB/T 4472 化工产品密度、相对密度的测定

GB/T 6678 化工产品采样总则

GB/T 6679 固体化工产品采样通则

GB/T 6680 液体化工产品采样通则

GB/T 8077 混凝土外加剂匀质性试验方法

GB 10238 油井水泥

GB/T 19139 油井水泥试验方法

SY/T 5377 钻井液参数测试仪器技术条件

3 术语和定义

下列术语和定义适用于本文件。

3.1

基准密度 datum density

供货商或用户提供的待测缓凝剂的密度。

3.2

适宜稠化时间范围 fitting thickening-time range

满足常规注水泥作业要求的合适的水泥浆稠化时间区域。

3.3

代表性温度点 typical temperature

对缓凝剂进行性能评价时采用的试验温度。

3.4

基准配方 basic recipe

在代表性温度点下满足稠化时间适宜范围的水泥浆配方。

3.5

空白配方 blank recipe

将基准配方中的缓凝剂除去后的配方。

3.6

稠化线形突变值 sudden change value of thickening-time

稠化试验开始 30min 后至稠度达到 40Bc 的时间段内,稠化曲线有鼓包和走台阶线形(指高于初始稠度并且持续 3min 以上的鼓包或台阶)的现象时,鼓包或台阶的峰值稠度与初始稠度的差值。

3.7

稠化时间延长值 extension value of thickening-time

空白配方与加入缓凝剂后的水泥浆稠化时间的差值。

3.8

加量敏感度 dosage sensitivity

同一配方,在推荐加量范围内,缓凝剂加量增加一定值,稠化时间的延长程度。

3.9

温度敏感度 temperature sensitivity

同一配方,在推荐加量范围内,温度增加一定值,稠化时间的缩短程度。

4 要求

4.1 物理性能

缓凝剂的外观、密度、水分、细度、水溶性的性能要求应符合表 1 的规定。

表 1 油井水泥缓凝剂物理性能要求

材料类别	检测项目	质量指标
固体类	外观	无受潮板结
	密度, g/cm ³	基准密度 ± 0.02
	水分, %	≤ 8
	细度 (0.315mm 筛余), %	< 15
	水溶性	溶解
液体类	外观	均匀无霉变
	密度, g/cm ³	基准密度 ± 0.02

4.2 应用性能

加有缓凝剂的基准配方(密度为 $1.90\text{g/cm}^3 \pm 0.01\text{g/cm}^3$)在其适用温度范围内的性能应符合表 2 的要求。

表 2 加有缓凝剂的基准配方水泥浆性能要求

检测项目	技术指标
初始稠度, Bc	≤ 30
过渡时间, min	≤ 40
稠化线形突变值, Bc	≤ 10
稠化时间延长值, min	> 60

表 2 (续)

检测项目	技术指标
加量敏感度, %	≤ 25
温度敏感度, %	≤ 20
游离液, %	≤ 1.4
24h 抗压强度, MPa	≥ 14

5 试验方法

5.1 仪器与材料

5.1.1 仪器

主要包括:

- 电子天平: 精度 0.01g。
- 恒温干燥箱: 温度范围 0℃~200℃。
- 干燥器: 内盛变色硅胶。
- 称量瓶: $\phi 25\text{mm} \times 65\text{mm}$ 。
- 比重瓶: 25mL 或者 50mL。
- 分析天平: 称量范围 0g~200g, 精度 0.1mg。
- 试验筛: 0.315mm 方孔筛。
- 恒速搅拌器、常压稠化仪、增压养护釜、抗压强度试验机、增压稠化仪: 应符合 GB 10238 的要求。
- 钻井液密度计: 应符合 SY/T 5377 的要求。
- 加压密度计、250mL 量筒: 应符合 GB/T 19139 的要求。

5.1.2 材料

主要包括:

- 油井水泥: 符合 GB 10238 要求的高抗硫酸盐型 (HSR) G 级水泥。
- 硅粉: 0.095mm~0.075mm (160 目~200 目) 筛余小于 4%, SiO_2 含量不低于 98%。
- 水: 蒸馏水。
- 缓凝剂: 待测样品。
- 与缓凝剂配伍的其他外加剂材料。

5.2 组批与采样

5.2.1 组批

5.2.1.1 生产商以一个生产批次进行组批。

5.2.1.2 油田用户以一次进货的产品量进行组批。

5.2.2 采样

5.2.2.1 按批为单位进行采样, 每批的采样数按 GB/T 6678 确定。

5.2.2.2 固体缓凝剂的采样按 GB/T 6679 进行。

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5.2.2.3 液体缓凝剂的采样按 GB/T 6680 进行。

5.2.2.4 固体缓凝剂的采样总量不少于 500g，液体缓凝剂的采样总量不少于 1000mL。

5.2.2.5 将按批采到的样品进行充分混匀后，等量分装于两个清洁、干燥的瓶中，密封并贴上标签。标签上应注明供货商名称、产品名称、批号和采样日期。两瓶样品中，一瓶用于评价，另一瓶留作仲裁用。留为仲裁的样品保存期应与样品保质期一致。

5.3 物理性能试验方法

5.3.1 外观测定

自然光条件下目测。

5.3.2 密度测定

固体缓凝剂按 GB/T 4472 中比重瓶法进行，液体缓凝剂按 GB/T 4472 中密度计法进行。

5.3.3 水分测定

固体缓凝剂的水分测定按 GB/T 8077 进行。

5.3.4 细度测定

固体缓凝剂的细度测定按 GB/T 8077 进行。

5.3.5 水溶性测定

将固体缓凝剂在 $23^{\circ}\text{C} \pm 1^{\circ}\text{C}$ 条件下配制成 5% 的溶液，用恒速搅拌器在 4000r/min 的转速下搅拌 50s 后，观察溶液底部有无沉淀或分层现象。

5.4 水泥浆性能试验方法

5.4.1 水泥浆基准配方的来源

水泥浆基准配方的来源主要包括：

- 用户检验：供货商提供的基准配方或与用户材料配套形成的基准配方。
- 供货商检验：供货商自行配套形成或根据用户要求组成的基准配方。
- 第三方检验：委托方提供的基准配方。

5.4.2 水泥浆制备

5.4.2.1 按照基准配方制备 $1.90\text{g}/\text{cm}^3 \pm 0.01\text{g}/\text{cm}^3$ 的水泥浆，制备方法按 GB/T 19139 进行。

5.4.2.2 水泥浆密度测定方法按 GB/T 19139 进行。

5.4.2.3 在 110°C (BHST) 以上温度试验时，应加入 35% (BWOC) 的硅粉。

5.4.3 水泥浆试验

5.4.3.1 代表性温度点的确定

代表性温度点依据该缓凝剂样品的适用温度范围确定，选取下限温度（低温）和上限温度（高温）为两端温度点，均匀分割取点，相临温度点之差最大不超过 25°C 。

5.4.3.2 稠化时间试验

以基准配方，在全部代表性温度点下，参考表 3 推荐的试验条件，按 GB/T 19139 进行稠化试

验，读取基准配方的初始稠度、过渡时间、稠化线形突变值及稠化时间。

表 3 水泥浆试验条件和适宜稠化时间

试验温度 (BHCT) ℃	试验压力 MPa	升温时间 min	稠化时间适宜范围 min	强度试验温度 (BHST) ℃
50	25.9	32	100~225	71
60	32.8	37	110~240	82
70	39.6	41	120~255	92
80	46.5	45	130~270	102
90	53.3	49	140~285	113
100	60.2	53	150~300	123
110	67.0	57	160~315	133
120	73.9	61	170~330	144
130	80.7	65	180~345	154
140	87.6	69	190~360	164
150	94.4	73	200~375	175
160	101.3	77	210~390	185
170	108.1	81	220~405	196
180	115.0	86	230~420	206

5.4.3.3 稠化时间延长值试验

- 根据现场需求，在某一代表性温度点下，测试空白配方的稠化时间。
- 在相同条件下，在空白配方中加入待测缓凝剂，测试稠化时间。
- 按公式 (1) 计算稠化时间延长值：

$$\Delta t = t - t_0 \quad \dots\dots\dots (1)$$

式中：

Δt ——稠化时间延长值，单位为分 (min)；

t ——加入缓凝剂后的稠化时间，单位为分 (min)；

t_0 ——空白配方的稠化时间，单位为分 (min)。

5.4.3.4 加量敏感度试验

根据现场需求，在某一代表性温度点下的基准配方，将缓凝剂加量增加 10% 进行稠化试验，读取稠化时间。按公式 (2) 计算缓凝剂加量敏感度：

$$R_{10} = \frac{t_{10} - t_p}{t_p} \times 100\% \quad \dots\dots\dots (2)$$

式中：

R_{10} ——缓凝剂加量敏感度，以百分数表示；

t_{10} ——增加 10% 加量条件下水泥浆的稠化时间，单位为分 (min)；

t_p ——基准配方的稠化时间，单位为分（min）。

5.4.3.5 温度敏感度试验

根据现场需求，以某一代表性温度点下的基准配方，将试验温度增加 5℃ 进行稠化试验，读取稠化时间。按公式（3）计算试验温度增加 5℃ 条件下的温度敏感度：

$$R_T = \frac{t_p - t_T}{t_p} \times 100\% \quad \dots\dots\dots (3)$$

式中：

R_T ——缓凝剂温度敏感度，以百分数表示；

t_p ——基准配方的稠化时间，单位为分（min）；

t_T ——增加 5℃ 后基准配方的稠化时间，单位为分（min）。

5.4.3.6 游离液试验

根据现场需求，以某一代表性温度点下的基准配方，按 GB/T 19139 进行。

5.4.3.7 抗压强度试验

根据现场需求，以某一代表性温度点下的基准配方，参考表 3 推荐的强度试验温度条件，按 GB/T 19139 进行。

6 评价报告

油井水泥缓凝剂的评价报告见附录 A。

附 录 A

(规范性附录)

油井水泥缓凝剂评价报告

样品名称: _____ 生产批号: _____ 生产厂商: _____
 适用温度范围: _____ 委托单位: _____ 送(采)样日期: _____
 送(采)样人: _____ 采样地点: _____ 评价单位: _____
 执行标准: _____ 评价用水: _____ 水泥批号: _____
 报告编号: _____

项目		指标	结果
固体类物理性能	外观	无受潮板结	
	密度, g/cm ³	基准密度 ± 0.02	
	水分, %	≤ 8	
	细度 (0.315mm 筛余), %	< 15	
	水溶性	溶解	
液体类物理性能	外观	均匀无霉变	
	密度, g/cm ³	基准密度 ± 0.02	
初始稠度, Bc		≤ 30	
过渡时间, min		≤ 40	
稠化线形突变值, Bc		≤ 10	
稠化时间延长值, min		> 60	
加量敏感度, %		≤ 25	
温度敏感度, %		≤ 20	
游离液, %		≤ 1.4	
24h 抗压强度, MPa		≥ 14	
备注		温 度: _____ °C 基准配方: _____ 稠化时间: _____ min	
结论			

实验人员: _____ 审核: _____ 批准: _____ 日期: _____

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**Evaluation method for well cement additives—
Part 1: Retarder**

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Annex A (Normative) Evaluation report of well cement retarder 18

Foreword

SY/T 5504 *Evaluation method for well cement additives* consists of eight parts;

- Part 1: *Retarder*;
- Part 2: *Fluid-loss additive*;
- Part 3: *Dispersant*;
- Part 4: *Accelerator*;
- Part 5: *Anti-gas migration agent*;
- Part 6: *Light-weight agent*;
- Part 7: *Weighting agent*;
- Part 8: *Expanding agent*.

This is Part 1 of SY/T 5504.

This part was drafted according to GB/T 1.1—2009 *Directives for standardization—Part 1: Structure and drafting of standards*.

This part replaces SY/T 5504.1—2005 *Evaluation method for well cement additives—Part 1: Retarder*. Compared with SY/T 5504.1—2005, the main changes of this part are as follows:

- Physical property table is added, and the performance requirements of pH value and solidifying point are deleted (4.1 in this part, 4.1 in SY/T 5504.1—2005);
- Requirements of sudden change value of consistency thickening-time linear sudden vibration value in application performance table are added (4.2 in this part);
- Index and test methods of thickening-time extension value are added (4.2 and 5.4.3.3 in this part);
- Test item, shape of thickening-time curve, is deleted (4.2.1 in SY/T 5504.1—2005);
- Test methods of pH value are deleted (5.3.6 in SY/T 5504.1—2005);
- Test methods of freezing point are deleted (5.3.7 in SY/T 5504.1—2005);
- Index and test methods of thickening-time adjustability are deleted (4.2.1 and 5.4.3.2 in SY/T 5504.1—2005);

—Index and test methods of temperature sensitivity are deleted (4.2.1 and 5.4.3.6 in SY/T 5504.1—2005);

Index and test methods of dosage sensitivity coefficient are deleted (4.2.1 and 5.4.3.7 in SY/T 5504.1—2005);

—Index and test methods of dosage sensitivity are added (4.2 and 5.4.3.4 in this part);

—Index and test methods of temperature sensitivity are added (4.2 and 5.4.3.5 in this part);

—Index and test methods of retarder quality stability are deleted (4.2.2 and 5.5 in SY/T 5504.1—2005);

—Determination methods of test temperature point are revised (5.4.3.1 in this part, 5.4.3.1 in SY/T 5504.1—2005);

—Evaluation report is revised (Annex A, Chapter 6 in SY/T 5504.1—2005).

Some contents in this document may be involved in patent. The publisher of this document assumes no responsibility of patent.

This part is proposed by and under the jurisdiction of China Petroleum Standardization Committee for Drilling Engineering.

This part was drafted responsibly by China Oil-field Services Limited.

This part was drafted secondly by SINOPEC Research Institute of Petroleum Engineering (SRIPE), Boxing Company of China National Petroleum Offshore Engineering Co. Ltd.

This part was mainly drafted by Yongsong Wang, Weikai Song, Tongyou Wang, Lianhao Duan, Wei Shen, Shidong Ding, Linhao Sha.

This part replaces SY/T 5504.1—2005.

The following previous editions of SY/T 5504.1—2005 are:

—SY/T 5504—1992; SY/T 5504—1996.

SY/T 5504.1—2013

This part is published in both Chinese and English texts, the Chinese version shall prevail. In the event of any discrepancy between the

Evaluation method for well cement additives— Part 1: Retarder

1 Scope

This part of SY/T 5504 specifies property requirements, evaluation methods and evaluation report for well cement retarder used in well cementing operations.

This part is applicable to evaluation of well cement retarder used in well cementing operations.

2 Normative references

The following referenced documents are essential to the application of this document. Only the dated editions of all the dated normative references are applicable to this document. The latest editions (including all revision forms) of all the undated normative references are applicable to this document.

GB/T 4472 *Determination of density and relative density for chemical products*

GB/T 6678 *General principles for sampling chemical products*

GB/T 6679 *General rules for sampling solid chemical products*

GB/T 6680 *General rules for sampling liquid chemical products*

GB/T 8077 *Methods for testing uniformity of concrete admixture*

GB 10238 *Oil well cement*

GB/T 19139 *Testing of well cements*

SY/T 5377 *Specifications for drilling fluid testing instruments*

3 Terms and definitions

The following terms and definitions are applica-

ble to this paper.

3.1 datum density

The density of retarder supplied by supplier or user.

3.2 fitting thickening-time range

The range of thickening-time suitable for conventional cementing operations.

3.3 typical temperature

The testing temperature adopted in the quality inspection for retarder.

3.4 basic recipe

The recipe of cement slurry, which meets the appropriate range of thickening-time at typical temperature.

3.5 blank recipe

The basic recipe without retarder.

3.6 sudden change value of thickening-time

The difference between the initial consistency and the top consistency of swell or step, from the time after starting thickening-time test for 30min to the time at which a consistency of 40 Bc is reached, when there is swell or step on consistency curve (the swell or step higher than initial consistency and maintaining for more than 3min).

3.7 extension value of thickening-time

The difference of thickening-time between the blank recipe and the recipe with retarder.

3.8 dosage sensitivity

The extending degree of thickening-time with

retarder dosage increase in a certain recipe and recommended dosage range.

3.9 temperature sensitivity

The shorten degree of thickening-time with the temperature increase in a certain recipe and recommended dosage range.

4 Requirements

4.1 Physical properties

The appearance, density, moisture, fineness, water-solubility of the retarder should meet the requirements of Table 1.

Table 1 Physical property requirements

Category	Item	Index
Solid	Appearance	No sweating and hardening
	Density, g/cm ³	Datum density ± 0.02
	Moisture, %	≤ 8
	Fineness (residue on 0.315mm sieve), %	< 15
	Water-solubility	Yes
Liquid	Appearance	Uniform and no mildew
	Density, g/cm ³	Datum density ± 0.02

4.2 Application properties

The properties of basic recipe (density is 1.90g/cm³ $\pm$ 0.01g/cm³) at typical temperature should meet the requirements of Table 2.

5 Test methods

5.1 Apparatus and materials

5.1.1 Apparatus

Mainly including:

- a) Electronic balance; With the precision of

0.01g.

- b) Constant temperature drying oven; With the temperature range of 0℃ to 200℃.
- c) Drier; Containing discolored silica gel.
- d) Weighing bottle; $\phi 25\text{mm} \times 65\text{mm}$.
- e) Density bottle; 25mL or 50mL.
- f) Analytical balance; With weighing range of 0g to 200g and the precision of 0.1mg.
- g) Sieve; With mesh size of 0.315mm.
- h) Slurry mixer, atmospheric consistometer, pressurized curing chamber, compressive strength tester, pressurized consistometer; Shall meet the stipulation of GB 10238.

Table 2 Property requirements for basic recipe cement slurry

Item	Index
Initial consistency, Bc	≤ 30
Transition time, min	≤ 40
Sudden change value of consistency, Bc	≤ 10
Extension value of thickening-time, min	> 60
Dosage sensitivity, %	≤ 25
Temperature sensitivity, %	≤ 20
Free fluid, %	≤ 1.4
Compressive strength for 24h, MPa	≥ 14

- i) Mud balance; Shall meet the stipulation of SY/T 5377.
- j) Pressurized density balance, 250mL graduated cylinder; Shall meet the stipulation of GB/T 19139.

5.1.2 Materials

Mainly including

- a) Well cement; Class G well cement (HSR), shall meet the stipulation of GB 10238.
- b) Silica flour; The residue on 0.075mm to 0.095mm (mesh 200 to 160) sieve is less than 4% and the SiO₂ content is no less than 98%.
- c) Water; Distilled water.
- d) Retarder; Test sample.
- e) Other additives; Compatible with the retarder.

5.2 Batching and sampling

5.2.1 Batching

5.2.1.1 Take one production circle as a batch unit for manufacturer.

5.2.1.2 Take one purchasing batch as a batch unit for user.

5.2.2 Sampling

5.2.2.1 Take a batch as a sampling unit, and determine the hits of each batch in accordance with GB/T 6678.

5.2.2.2 Obtain solid sample in accordance with GB/T 6679.

5.2.2.3 Obtain liquid sample in accordance with GB/T 6680.

5.2.2.4 The amount of solid sample should be no less than 500g, and the amount of liquid sample should be no less than 1000mL.

5.2.2.5 All samples obtained from the same batch should be well mixed, evenly stored in two clean and dry bottles, and sealed. Each bottle shall be labeled and identified with the supplier's name, product name, batch number and

date of sampling. One of the two bottles is for evaluation, and the other is for arbitration. Samples for arbitration and reference should have the same guarantee period.

5.3 Test procedures for physical properties

5.3.1 Appearance test

Performed by visual observation under the natural light.

5.3.2 Density test

Performed in accordance with GB/T 4472.

5.3.3 Moisture test

Performed in accordance with GB/T 8077.

5.3.4 Fineness test

Performed in accordance with GB/T 8077.

5.3.5 Water-solubility test

Performed at 23℃ ± 1℃ as follows; Prepare retarder solution of 5% concentration, stir for 50s at 4000 r/min with slurry mixer, and then observe if sedimentation or stratification appears.

5.4 Test procedures for cement slurry properties

5.4.1 Source of basic recipe

Source of basic cement slurry recipe mainly includes;

——User test; The recipe provided by the supplier or the recipe formed by the user with his own compatible materials.

——Supplier test; The recipe provided by the supplier or on the request of the user.

——Third party test; The recipe provided by the client.

5.4.2 Preparation of cement slurry

5.4.2.1 Prepare basic recipe slurry of 1.90g/cm³ ± 0.01g/cm³ in accordance with GB/T 19139.

5.4.2.2 Density test should be performed in accordance with GB/T 19139.

5.4.2.3 35% (BWO) silica flour should be added when the test temperature is at 110℃ (BHST) or higher.

5.4.3 Cement slurry test

5.4.3.1 Determination of the typical temperatures

The typical temperature should be determined by the applicable temperature range of the retarder. It is evenly selected between the lower temperature limit and the upper temperature limit, and the temperature interval should be no more than 25℃.

5.4.3.2 Thickening-time test

Referring to the conditions listed in Table 3, thickening-time test should be carried out at all typical temperatures in accordance with GB/T 19139, and then the initial consistency, transition time, sudden change value of consistency

and thickening-time of the basic recipe should be read and recorded.

5.4.3.3 Test of thickening-time extension value

- Test the thickening-time of blank recipe at a typical temperature according to the site requirement.
- Test the thickening-time of recipe with retarder, under the same conditions.
- Calculate the extension value of thickening-time according to Equation (1):

$$\Delta t = t - t_0 \quad \dots\dots\dots (1)$$

Where:

Δt is the extension value of thickening-time, min;

t is the thickening-time of recipe with retarder, min;

t_0 is the thickening-time of blank recipe, min.

Table 3 Test condition and fitting thickening-time of cement slurry

Test temperature (BHCT) ℃	Test pressure MPa	Time to BHCT min	Fitting thickening- time range min	Temperature for strength test (BHST) ℃
50	25.9	32	100~225	71
60	32.8	37	110~240	82
70	39.6	41	120~255	92
80	46.5	45	130~270	102
90	53.3	49	140~285	113
100	60.2	53	150~300	123
110	67.0	57	160~315	133
120	73.9	61	170~330	144
130	80.7	65	180~345	154
140	87.6	69	190~360	164
150	94.4	73	200~375	175
160	101.3	77	210~390	185
170	108.1	81	220~405	195
180	115.0	86	230~420	205

5.4.3.4 Dosage sensitivity test

Prepare cement slurry according to the basic recipe, then carry out thickening-time test by increasing retarder dosage by 10% at a typical temperature according to the site requirement, and record the result. Calculate the dosage sensitivity of retarder according to Equation (2):

$$R_D = \frac{t_D - t_p}{t_p} \times 100\% \quad \dots\dots (2)$$

Where:

R_D is the dosage sensitivity of retarder, in percent;

t_D is the thickening-time of cement slurry by increasing retarder dosage by 10%, min;

t_p is the thickening-time of basic recipe, min.

5.4.3.5 Temperature sensitivity test

Prepare cement slurry according to the basic recipe of a typical temperature according to the site requirement, then carry out thickening-time test by increasing the temperature by 5℃, and record the result. Calculate the temperature sensitivity according to Equation (3):

$$R_T = \frac{t_p - t_T}{t_p} \times 100\% \quad \dots\dots (3)$$

Where:

R_T is the temperature sensitivity of retarder, in percent;

t_p is the thickening-time of basic recipe, min.

t_T is the thickening-time of basic recipe by increasing the temperature by 5℃, min.

5.4.3.6 Free fluid test

Prepare cement slurry according to the basic recipe, and then carry out free fluid test at a typical temperature in accordance with GB/T 19139.

5.4.3.7 Compressive strength test

Prepare cement slurry according to the basic recipe, and then carry out compressive strength test at a typical temperature referring to the conditions listed in Table 3, and in accordance with GB/T 19139.

6 Evaluation report

See Annex A for evaluation report of well cement retarder.

Annex A

(Normative)

Evaluation report of well cement retarder

Product name: _____

Batch number: _____

Supplier: _____

Applicable temperature range: _____

Client: _____

Date of sampling: _____

Person of sampling: _____

Sampling place: _____

Evaluation unit: _____

Executive standard: _____

Water for evaluation: _____

Batch number of cement: _____

Report number: _____

Item		Index	Result
Physical properties of solid	Appearance	No sweating and hardening	
	Density, g/cm ³	Datum density ± 0.02	
	Moisture, %	≤ 8	
	Fineness (residue on 0.315mm sieve), %	< 15	
	Water-solubility	Yes	
Physical properties of liquid	Appearance	Uniform and no mildew	
	Density, g/cm ³	Datum density ± 0.02	
Initial consistency, Bc		≤ 30	
Transition time, min		≤ 40	
Sudden change value of consistency, Bc		≤ 10	
Extension value of thickening-time, min		> 60	
Dosage sensitivity, %		≤ 25	
Temperature sensitivity, %		≤ 20	
Free fluid, %		≤ 1.4	
Compressive strength for 24h, MPa		≥ 14	
Note		Temperature: ____ °C (Basic recipe: _____ Thickening-time: _____ min)	
Conclusion			

Experimenter: _____

Auditor: _____

Approver: _____

Date: _____